

# Ethereum-based smart contract implementation

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## 1. Token Creation (ERC20)

```
//Owner can mint token to any address, only the owner have this functionality
function mint(address to, uint256 amount) public onlyOwner {
    _mint(to, amount);
}

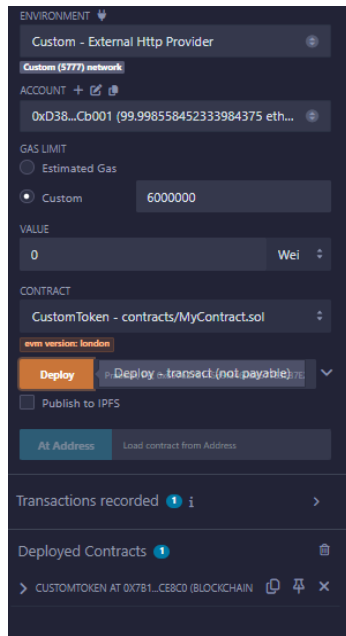
//token transfer can be paused or unpaused by owner
function pause() public onlyOwner {
    _pause();
}

function unpause() public onlyOwner {
    _unpause();
}

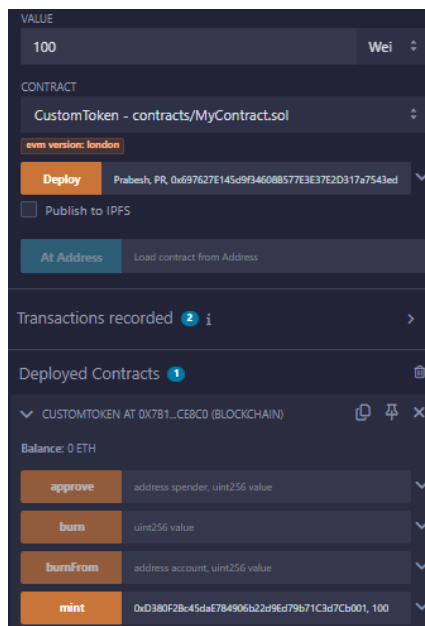
// Multiple inheritance so overriding _update method and specifying which parent's _update
function _update(address from, address to, uint256 value) internal override(ERC20, ERC223) {
    super._update(from, to, value);
}
```

Code snippet

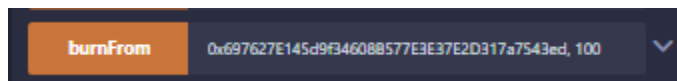
This is a simple code and all we used was functions from the library to create tokens. Below are the functionality in detail.



## Initial deployment



The owner can mint token to any address, only the owner have this functionality



## Burning

There is also a pause and unpause feature from the library implemented.

## 2. Voting System

The screenshot shows a web application for a voting system. The interface is divided into several sections:

- DEPLOY & RUN TRANSACTIONS:** This section shows the deployed contracts and their details. It includes a table with columns for the contract name, address, and a 'Transact' button.
- Low level interactions:** This section shows the low-level interactions with the contracts. It includes a table with columns for the contract name, address, and a 'Transact' button.
- Code Editor:** The right panel displays the Solidity code for the voting system. The code is syntax-highlighted and includes comments. The code defines a `VotingSystem` contract with functions for adding candidates, registering voters, and voting.

### Overall voting system

The screenshot shows a transaction log with two entries:

- [block:59 txIndex:-] from: 0xa65...c0f70 to: VotingSystem.addCandidate(string) 0x0b8...8e2c7 value: 0 wei data: 0x462...00000 logs: 1 hash: 0x76b...ad61f  
transact to VotingSystem.addCandidate pending ...
- [block:60 txIndex:-] from: 0xa65...c0f70 to: VotingSystem.addCandidate(string) 0x0b8...8e2c7 value: 0 wei data: 0x462...00000 logs: 1 hash: 0x5fc...713df  
transact to VotingSystem.addCandidate pending ...

### Adding candidates

The screenshot shows a transaction log with one entry:

- [block:61 txIndex:-] from: 0xa65...c0f70 to: VotingSystem.registerVoter(address) 0x0b8...8e2c7 value: 0 wei data: 0x38d...cb001 logs: 1 hash: 0x8ec...42066  
transact to VotingSystem.registerVoter pending ...

### Registering voters

The screenshot shows a transaction log with one entry:

- [block:62 txIndex:-] from: 0xd38...cb001 to: VotingSystem.vote(uint256) 0x0b8...8e2c7 value: 0 wei data: 0x012...00000 logs: 1 hash: 0xfa1...f42d1  
transact to VotingSystem.vote pending ...

### Voting for a candidate

The screenshot shows a transaction log with one entry:

- [block:63 txIndex:-] from: 0xa65...c0f70 to: VotingSystem.vote(uint256) 0x0b8...8e2c7 value: 0 wei data: 0x012...00000 logs: 1 hash: 0x76b...ad61f  
transact to VotingSystem.vote pending ...  
Error while querying the provider: VM Exception while processing transaction: revert Only registered voters ha permission

### Unregistered voter trying to vote

The screenshot shows a transaction log with one entry:

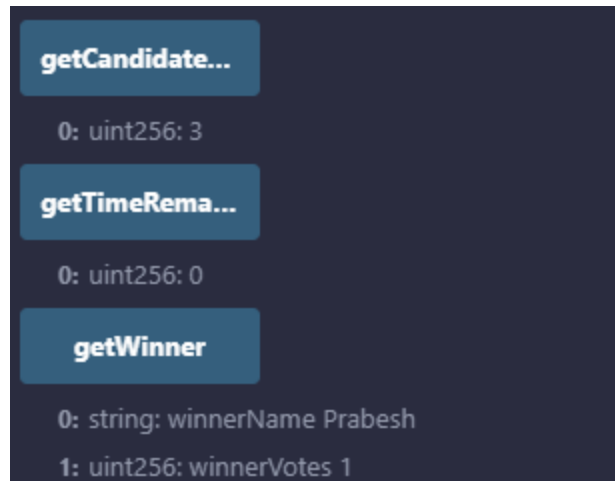
- [block:64 txIndex:-] from: 0xa65...c0f70 to: VotingSystem.vote(uint256) 0x0b8...8e2c7 value: 0 wei data: 0x012...00000 logs: 1 hash: 0x76b...ad61f  
transact to VotingSystem.vote pending ...  
Error while querying the provider: VM Exception while processing transaction: revert You have already voted

### Trying multiple votes

The screenshot shows a transaction log with one entry:

- [block:65 txIndex:-] from: 0xa65...c0f70 to: VotingSystem.vote(uint256) 0x0b8...8e2c7 value: 0 wei data: 0x012...00000 logs: 1 hash: 0x76b...ad61f  
transact to VotingSystem.vote pending ...  
Error while querying the provider: VM Exception while processing transaction: revert Voting has not ended yet

### The winner can't be determined until the time is up



Winner shown in the frontend

### 3. Escrow system

```
//Buyer sends ETH to this function
function depositFunds(uint256 transactionId) public payable {  ⚡ infinite gas
    Transaction storage transaction = transactions[transactionId];

    require(msg.sender == transaction.buyer, "Only buyer can deposit");
    require(transaction.state == State.AWAITING_DEPOSIT, "Invalid state");
    require(msg.value > 0, "Deposit amount must be greater than 0");

    transaction.amount = msg.value;
    transaction.state = State.AWAITING_DELIVERY; // Update the status to await delivery

    emit FundsDeposited(transactionId, msg.value);
}

//When buyer approves delivery of product, funds are sent to seller
function approveDelivery(uint256 transactionId) public {  ⚡ infinite gas
    Transaction storage transaction = transactions[transactionId];

    require(msg.sender == transaction.buyer, "Only buyer can approve delivery");
    require(transaction.state == State.AWAITING_DELIVERY, "Invalid transaction state");

    transaction.buyerApproved = true;
    _releaseFunds(transactionId);

    emit DeliveryApproved(transactionId);
}

//Buyer can raise dispute if they didnt receive the product, Now the third person is involved
function raiseDispute(uint256 transactionId) public {  ⚡ infinite gas
```

Code snippet of escrow system

|                  |                                         |   |
|------------------|-----------------------------------------|---|
| approveDeliv...  | uint256 transactionId                   | ▼ |
| createTransac... | 0x697627E145d9f346088577E3E37E2D317     | ▼ |
| depositFunds     | uint256 transactionId                   | ▼ |
| raiseDispute     | uint256 transactionId                   | ▼ |
| resolveDispute   | uint256 transactionId, uint8 resolution | ▼ |

Buyer creates a transaction with the seller id

|              |                                     |
|--------------|-------------------------------------|
| depositFunds | 0 depositFunds - transact (payable) |
|--------------|-------------------------------------|

Buyer deposits to transaction id 0

|              |                                         |
|--------------|-----------------------------------------|
| raiseDispute | 0 raiseDispute - transact (not payable) |
|--------------|-----------------------------------------|

Buyer raises dispute to transaction id 0

|            |                                                        |
|------------|--------------------------------------------------------|
| arbitrator | 0: address: 0xa658989a8b0b718c85281E61F285ec22CbEC0F70 |
|------------|--------------------------------------------------------|

Arbitrator id

|                |                                                        |
|----------------|--------------------------------------------------------|
| resolveDispute | 0,1                                                    |
| arbitrator     | 0: address: 0xa658989a8b0b718c85281E61F285ec22CbEC0F70 |

Arbitrator resolves dispute

```

[block:109 txIndex:-] from: 0xa65...c0f70 to: EscrowService.(constructor) value: 0 wei data: 0x608...a0033
logs: 0 hash: 0x816...12899
transact to EscrowService.createTransaction pending ...

[block:110 txIndex:-] from: 0xa65...c0f70 to: EscrowService.createTransaction(address) 0xf17...da106
value: 0 wei data: 0xa23...543ed logs: 1 hash: 0x4de...1d612
transact to EscrowService.depositFunds pending ...

[block:111 txIndex:-] from: 0xa65...c0f70 to: EscrowService.depositFunds(uint256) 0xf17...da106
value: 19 wei data: 0x3b7...00000 logs: 1 hash: 0xe7b...20cfc
transact to EscrowService.approveDelivery pending ...

[block:112 txIndex:-] from: 0xa65...c0f70 to: EscrowService.approveDelivery(uint256) 0xf17...da106
value: 0 wei data: 0xc94...00000 logs: 2 hash: 0xde6...b761f

```

Approving delivery without disputes

## 4. Decentralized auction

```
// Starting the auction with item description, starting bid and minimum increment requ.
function startAuction( /* infinite gas
    string memory _itemDescription,
    uint256 _startingBid,
    uint256 _minBidIncrementPercent
) public onlyOwner {
    require(!auctionStarted, "Auction already started");
    require(!auctionEnded, "Auction already ended");

    itemDescription = _itemDescription;
    highestBid = _startingBid;
    minBidIncrement = _minBidIncrementPercent;
    auctionStarted = true;

    emit AuctionStarted();
}

//Bid must be higher than current highest bid and increment
function bid() public payable auctionIsActive { /* infinite gas
    uint256 minRequiredBid = highestBid + ((highestBid * minBidIncrement) / 100);
    require(msg.value >= minRequiredBid, "Bid not high enough");

    if (highestBidder != address(0)) {
        pendingReturns[highestBidder] += highestBid;
    }

    highestBidder = msg.sender;
    highestBid = msg.value;
}
```

Auction code snippet

ENGLISHAUCTION AT 0xE10

Balance: 0.0000000000000000017 ETH

bid

claimFunds

claimItem

endAuction

startAuction

\_itemDescription: Prabesh

\_startingBid: 5

\_minBidIncrementPercent: 1

Calldata Parameters transact

withdraw

Initial UI of Auction system, the failed bidders can withdraw their amount with withdraw.

```
[block:117 txIndex:-] from: 0xa65...c0f70 to: EnglishAuction.(constructor) value: 0 wei
data: 0x608...a0033 logs: 0 hash: 0x1f6...64ac2
creation of EnglishAuction pending...

[block:118 txIndex:-] from: 0xa65...c0f70 to: EnglishAuction.(constructor) value: 0 wei
data: 0x608...a0033 logs: 0 hash: 0x43a...ae7ce
transact to EnglishAuction.startAuction pending ...

[block:119 txIndex:-] from: 0xa65...c0f70
to: EnglishAuction.startAuction(string,uint256,uint256) 0xe10...fca46 value: 0 wei data: 0x2ac...00000
logs: 1 hash: 0x389...3a4e0
transact to EnglishAuction.bid pending ...

[block:120 txIndex:-] from: 0xd38...cb001 to: EnglishAuction.bid() 0xe10...fca46 value: 7 wei
data: 0x199...8aeeef logs: 1 hash: 0x128...670bb
transact to EnglishAuction.bid pending ...

[block:121 txIndex:-] from: 0x697...543ed to: EnglishAuction.bid() 0xe10...fca46 value: 10 wei
data: 0x199...8aeeef logs: 1 hash: 0x2af...6216a
transact to EnglishAuction.bid pending ...
```

Starting auction and 3 ids bidding

```
transact to EnglishAuction.endAuction pending ...

Error while querying the provider: VM Exception while processing transaction: revert Only the auction owner can call this function
transact to EnglishAuction.endAuction errored: processing response error (body="{\"id\":379,\"jsonrpc\":\"2.0\",\"error\":{\"message\":\"VM
transact to EnglishAuction.endAuction pending ...

[block:123 txIndex:-] from: 0xa65...c0f70 to: EnglishAuction.endAuction() 0xe10...fca46 value: 0 wei
data: 0xfe6...7a54b logs: 1 hash: 0xa01...b48c8
transact to EnglishAuction.claimFunds pending ...

[block:124 txIndex:-] from: 0xa65...c0f70 to: EnglishAuction.claimFunds() 0xe10...fca46 value: 0 wei
data: 0xac3...07773 logs: 1 hash: 0x6e0...268d4
transact to EnglishAuction.claimItem pending ...

Error while querying the provider: VM Exception while processing transaction: revert Only the highest bidder can claim the item
transact to EnglishAuction.claimItem errored: processing response error (body="{\"id\":448,\"jsonrpc\":\"2.0\",\"error\":{\"message\":\"VM
transact to EnglishAuction.claimItem pending ...

[block:125 txIndex:-] from: 0x0a0...391c0 to: EnglishAuction.claimItem() 0xe10...fca46 value: 0 wei
data: 0x154...d3126 logs: 1 hash: 0x58c...92440
```

Trying to claim item and funds from different ids and also the corrent id

getAuctionD...

```
0: string: item Prabesh
1: address: leading 0x0A07271d245509bcf
  88D53763C3167999d6391c0
2: uint256: topBid 15
3: bool: started true
4: bool: ended true
5: bool: claimed true
```

Auction details

The code details are provided in the sol file with easy comments for documentation.